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MILITARY SPECIFICATION

GENERATOR SET, GAS TURBINE, MAGNETIC

MINESWEEPING, 1750 KILOWATT

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a 1750-kilowatt (kw) impulse type, direct current (d.c.) minesweeping generator set and associated control equipment (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

FF-B-171 - Bearings, Ball, Annular (General Purpose).
FF-B-185 - Bearings, Roller, Cylindrical; and Bearings, Roller, Self-Aligning.
TT-P-645 - Primer, Paint, Zinc-Chromate, Alkyd Type.

MILITARY

MIL-C-104 - Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116 - Preservation-Packaging, Methods of.
MIL-B-117 - Bags, Sleeves and Tubing - Interior Packaging.
MIL-R-196 - Repair Parts for Internal Combustion Engines, Packaging of.
MIL-B-233 - Boxes, Repair Parts, Storage.
MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-A-907 - Antiseize Compound, High Temperature.
MIL-E-917 - Electric Power Equipment, Basic Requirements for (Naval Shipboard Use).
MIL-M-2130 - Motor-Generators, General Purpose (Naval Shipboard Use).
MIL-C-2212 - Controllers, Electric Motor, A.C. or D.C. and Associated Switching Devices, Naval Shipboard.
MIL-G-3111 - Generators, Electric, Direct-Current (Naval Shipboard Use).
MIL-E-5607 - Engine, Gas Turbine, Preparation for Storage and Shipment of, Process for.
MIL-A-8625 - Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-P-15024 - Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5 - Plates, Identification.
MIL-E-15090 - Enamel, Equipment, Light-Gray (Formula No. 111).
MIL-P-15328 - Primer (Wash) Pretreatment, Blue (Formula No. 117-B For Metals)
MIL-L-15719 - Lubricating Grease (High-Temperature, Electric Motor, Ball and Roller Bearings).
MIL-C-15730 - Coolers, Fluid, Industrial, Naval Shipboard: Lubricating Oil, Hydraulic Oil, and Fresh Water.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Ship Engineering Center, SEC 6124, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1075

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MILITARY (continued)

MIL-M-16034 - Meters, Electrical-Indicating (Switchboard and Portable Types)
MIL-I-16165 - Interference Shielding, Engine Electrical Systems.
MIL-C-16173 - Corrosion Preventive Compound, Solvent Cutback, Cold Application.
MIL-E-16298 - Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging of.
MIL-P-17286 - Propulsion and Auxiliary Steam Turbines and Gears (Including Repair Parts, Tools, Accessories and Instruments); Packaging of.
MIL-E-17341 - Engines, Gas Turbine, Propulsion and Auxiliary, Naval Shipboard.
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts: Packaging and Packing of.
MIL-F-18866 - Fittings, Flared Tube, 37-Degree, Steel.
MIL-C-19836 - Coolers, Fluid, Industrial, Air, Motor and Generator, Naval Shipboard.
MIL-F-20670 - Flanges, Pipe, Carbon Steel 150 P.S.I., W.S.P. (For Naval Shipboard Use).
MIL-M-24365 - Maintenance Engineering Analysis: Establishment of, and Procedures and Formats for Associated Documentation; General Specification for.
DOD-G-24508 - Grease, High Performance, Multi-Purpose (METRIC).

STANDARDS

MILITARY

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-242 - Electronic Equipment Parts (Selected Standards).
MIL-STD-278 - Fabrication Welding and Inspection; and Casting Inspection and Repair For Machinery, Piping and Pressure Vessels in Ships of the United States Navy.
MIL-STD-471 - Maintainability Demonstration.
MIL-STD-721 - Definitions of Effectiveness Terms for Reliability, Maintainability, Human Factors, and Safety.
MIL-STD-749 - Preparation and Submission of Data for Approval of Nonstandard Parts.
MIL-STD-781 - Reliability Tests: Exponential Distribution.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing with Appropriate Test Methods.
MIL-STD-1552 - Provisioning Technical Documentation, Uniform DOD Requirements for.
MIL-STD-1561 - Provisioning Procedures, Uniform DOD.
MS17830 - Nut, Self-Locking, Hexagon-Regular Height, 250°F (Non-Metallic Insert) 300 Series Corrosion Resistant Steel.

PUBLICATION

MILITARY

NAVSHIPS 0943-015-6010 - Sliding Surface Bearing Standardization Manual.
NAVSEA 0981-LP-052-8140 - Electrical Equipment Small Tray Magnetic Field Manual for Design of.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules, and Regulations.

(Application for copies should be addressed to the Uniform Classification Committee, Agent, G. F. Earl, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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NATIONAL CLASSIFICATION BOARD
National Motor Freight Classification Classes and Rules.

(Application for copies should be addressed to the ATA Tariff Section, 1616 P Street, N.W., Washington, DC 20036.)

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Generator set and associated control equipment. Generator set shall consist of a gas turbine engine geared to a d.c. generator and a fly wheel all mounted on a common subbase. Associated control equipment shall consist of a motor-generator exciter set, switchboard, an auxiliary controller, contactor panel, transfer panel, synchronizer signal source, and necessary appurtenances to constitute a complete electrical system for the generation and control of the magnetic minesweeping power specified herein.

3.2 General requirements.

3.2.1 Limitations.

3.2.1.1 Size and weight. Generator set complete with subbase and attached accessories shall not exceed the following:

Length - 12 feet (ft) 2 inches (in)
Width - 6 ft 6 in
Height - 7 ft 0 in
Weight - 25,000 pounds (lbs)

3.2.1.2 Magnetic material limitations. Magnetic materials shall not be used except for necessary magnetic circuits of electrical equipment. Nonmagnetic materials are defined as those materials having a permeability of less than 1.2 before fabrication and not greater than 2.0 after fabrication as measured in a 200 Oersted field.

3.2.1.3 Stray magnetic field limitations. Electric circuits shall be designed for self compensation of stray magnetic fields. Information on minimizing stray magnetic fields is contained in NAVSEA 0981-LP-052-8140. Numerical values of stray magnetic field limits are given in a classified amendment to this specification.

3.2.1.4 Eddy current field limitations. To limit magnetic fields due to eddy currents induced by movement in the earth's field, the subbase and large components shall be constructed of low conductivity material or shall be assembled with inserts of nonconducting material, so arranged as to limit the size of any closed loop area to 5 square feet (ft^2). Major components shall be insulated from each other to avoid loop areas greater than 5 ft^2 unless they are manufactured of low conductivity material. Low conductivity material is defined as material having a conductivity not greater than 10 percent of that of copper. In no case shall loop areas exceed 100 ft^2. Fuel and oil lines and armored cable shall be isolated from the subbase or shall have nonmetallic inserts so arranged that insulating inserts are not short circuited.

3.2.2 Shock resistance. Equipment shall meet the grade A, class I, type A requirements of MIL-S-901.

3.2.3 Material. Material shall be such as to insure continued satisfactory operation under the extreme conditions of heat, moisture, vibration, and shock encountered onboard ship. Reclaimed materials shall be used to the maximum extent possible.

3.2.4 Lifting gear. Heavy pieces of equipment shall be provided with nonmagnetic eyebolts and tapped holes, or other means satisfactory to the command or agency concerned as are required for lifting and handling.

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3.2.5 Inclined operation. Unit shall operate in accordance with the requirements herein when inclined permanently up to 15 degrees to either side or 5 degrees on either end from the normal horizontal position and with a momentary inclination of 45 degrees to either side simulating the roll of the ship, or by momentary end tilting of 10 degrees. Momentary herein means for a period up to 10 seconds.

3.2.6 Painting. Surfaces of metallic components, except working surfaces of machined parts, machined surfaces of couplings, shafts and similar items manufactured of corrosion-resistant material and high temperature parts of gas turbines shall be given one pretreatment coating in accordance with MIL-P-15328 and one coat of primer in accordance with TT-P-645 prior to assembly. After assembly, external surfaces shall be given two finish coats of enamel in accordance with class 2, type 1 of MIL-E-15090.

3.2.7 Welding. Welding shall be in accordance with MIL-STD-278.

3.2.7.1 Where equipment uses silver-brazed piping, face feed fittings may be used only in sizes below 1/2 inch. Only preinserted ring fittings shall be used in sizes 1/2 inch and above.

3.2.8 Protection against corrosion. In order to prevent deterioration due to corrosion, bolts, nuts, studs, pins, springs, screws, cap screws, and other fastenings or fittings shall be of a corrosion-resisting material as defined in MIL-E-917, or of a material treated in a manner to render it resistant to corrosion. Unpainted surfaces of aluminum or aluminum alloys shall be treated in accordance with the requirements of MIL-A-8625. Contact between dissimilar aluminum alloys or dissimilar metals shall be avoided. Where assembly of dissimilar metals with aluminum alloy is unavoidable, the alloys shall have their faying surfaces anodized. Where aluminum threaded surfaces or faying surfaces may require disassembly in service, an antiseize compound in accordance with MIL-A-907 shall be used. For the assembly of aluminum alloy parts, through-bolting is preferred, this applies with particular emphasis to comparatively high stressed assemblies or those subject to vibrating loads. Where the use of tap-bolts, stud-bolts, or machine screws is necessary, these shall be threaded into corrosion-resisting steel inserts cast or screwed into the aluminum alloy. Inserts shall be pinned, prick punched, or otherwise secured in such a manner as to prevent their backing out. No alloys of copper such as brass or bronze will be acceptable for use in threaded contact with aluminum alloys. The use of magnesium and magnesium alloys is prohibited.

3.2.8.1 Sea water cooled heat exchangers. Sea water cooled heat exchangers for the generator shall be class 2 in accordance with MIL-C-19836.

3.2.9 Identification and information plates. Identification plates of corrosion-resisting steel or nickel-copper alloy in accordance with type B or C of MIL-P-15024 and MIL-P-15024/5 shall be furnished for major items of equipment and shall show the NAVSEA number of the equipment service manual. Information plates shall be furnished in accordance with type E of MIL-P-15024 and MIL-P-15024/5 (clear acrylic plastic may be used) unless information plate location temperature is in excess of that for plastics, then type B will be required.

3.2.10 Ball and roller bearings. Ball and roller bearings shall be in accordance with FF-B-171 or FF-B-185. Drawings, specifications, manuals, and repair part cartons shall describe completely the bearing manufacturer's part designation and its appropriate National stock number. For the generator portion of the set, double-sealed bearings, prelubricated with grease conforming to DOD-G-24508, are preferred. Use grease conforming to MIL-L-15719 where bearing temperature exceeds 149°C. Where the temperature of the outer race of bearings exceeds 110°C (ambient plus heat rise) the bearings shall be heat stabilized to operate at higher than normal temperatures. Criteria for heat stabilization will be such that the bearing operating temperature will not cause the dimensions to increase as much as 0.0001 inch per inch of diameter when held at a temperature of 300°F for 2500 hours.

3.2.11 Electrical equipment basic requirements. Unless otherwise specified herein, the basic requirements of electrical equipment shall be in accordance with MIL-E-917.

3.2.12 Mechanical vibration. Vibration sensitive equipment shall meet the mechanical vibration limits of MIL-STD-167-1, type I. The complete gas turbine generator set mounted on the subbase shall meet the mechanical vibration limits of MIL-STD-167-1, type II.

3.2.13 Threads. Threads shall be in accordance with the requirements of Handbook H28. Taper pipe thread connections shall not be used. Fluid connections shall be direct brazed or welded, flanged, or by fittings in accordance with MIL-F-18866.

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3.2.14 Self-locking nuts. Self-locking nuts conforming to MS17830 shall be used with bolts for mounting components to the subbase.

3.3 Generator. Generator shall be in general accordance with MIL-G-3111, except as otherwise specified herein.

3.3.1 Rating. Generator shall deliver 1750 kW, 5000 amperes (A), 350 volts (V) on a square wave pulse of 5-1/2 seconds "ON" and 9-1/2 seconds "OFF". Generators shall be capable of producing rated output throughout the design speed range of the engine-generator set.

3.3.2 Regulation. When working into a resistive load the generator shall build up to 100 percent of any selected square wave pulse current within the generator rating in not more than 0.8 second without overshoot of more than 5 percent of the selected value and by the end of the first 1.0 seconds of the pulse, measured from the start of build up, shall have returned to within a band of plus or minus 2.5 percent of the selected value and shall remain within this band for the remainder of the pulse. At the end of a pulse, generator current shall decay to zero in not more than 0.7 second, and shall maintain zero output plus or minus 50 A for the duration of "OFF" period except that overshoot of 5 percent of the previous pulse current may occur during the first 1.0 second of the "OFF" period. If a double armature generator is used, current balance between the two armatures shall be such that the current load of one armature shall not differ from that armature's proportionate share of the total current load by more than 5.0 percent. This balance shall be obtained under any condition of load or temperature without manual adjustment.

3.3.3 Enclosure. Generators shall be totally enclosed, with built in air-to water heat exchangers designed to provide the cooling required for the specified class of insulation using sea water at a temperature of 85°F. Thermometers shall be installed in the air stream on the inlet and outlet sides of the coolers. Cooling water shall be supplied by a sea-water pump (see 3.6.7). No silicone materials shall be used inside the generator enclosure.

3.3.4 Ventilation. Generators shall be self-ventilated. Air filters shall be provided.

3.3.5 Insulation. Insulation shall be class B of MIL-E-917. Insulation materials used shall be compatible with the engine and generator bearing lubricating oils (see 3.6.8) that is, oil or oil vapors coming in contact with the generator insulation shall not soften or cause deterioration of the insulation, thereby causing premature generator failure.

3.3.6 Bearings. Bearings may be self-alining sleeve bearings or anti-friction bearings (see 3.2.10). If sleeve bearings are used, lubrication shall be by oil supplied by the engine lubrication system and a contact maker shall be provided for a high temperature alarm. Sleeve bearings shall be designed in accordance with NAVSHIPS 0943-015-6010.

3.3.7 Overspeed. Generator shall withstand a speed of 115 percent maximum rated speed (see 4.9).

3.3.8 Stray magnetic fields. Design and construction techniques for generators shall be implemented to avoid generation of stray magnetic fields. Information on minimizing stray magnetic fields is contained in NAVSEA 0981-LP-052-8140. In addition, the requirements in 3.3.8.1 through 3.3.8.7 shall be implemented.

3.3.8.1 Magnet frames of generator shall be of one piece construction (or laminated construction within a one-piece supporting frame), machined inside and outside to insure uniform cross section throughout. Frame weld shall be located on the centerline of the top main field pole.

3.3.8.2 Protuberances of magnetic material shall be avoided.

3.3.8.3 Generator stators shall have at least 12 field poles and the same number of interpoles.

3.3.8.4 Pole face windings shall have an even number of slots per pole face.

3.3.8.5 Commutators shall have three brush collector rings with the center ring carrying full commutator current and outside rings carrying half current.

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3.3.8.6 Each set of collector rings shall connect to three or four terminals arranged in accordance with NAVSEA 0981-LP-052-8140. Terminals shall be located above the horizontal centerline of the generator.

3.3.8.7 For generators of double armature construction, field poles of "N" polarity for one armature shall be opposite field poles of "S" polarity on the other.

3.3.9 Internal illumination. Lighting fixtures shall be installed for internal illumination of the commutator and brushes. Supply voltage shall be 120 V alternating current (a.c.). Control switches shall be provided on the generator housings.

3.3.10 Space heaters. Space heaters shall be provided to operate from 450 V a.c., 3-phase and shall be arranged to provide a balanced load.

3.3.11 Terminals. Terminals shall be silver-plated and shall be drilled for terminal lugs which shall be furnished with each generator.

3.3.12 Vibration. Generators shall have no critical speeds between 65 and 115 percent of nominal speed.

3.3.13 Bolted connections. Contact surfaces of bolted connections which carry load current shall be silver-plated.

3.4 Exciter set. Exciter set may be a motor-generator set or may be driven as an accessory.

3.4.1 Motor-generator exciter. Motor-generator exciter shall be in accordance with MIL-M-2130, except as otherwise specified herein. The set shall consist of a 440 V, 60-hertz (Hz), 3-phase induction motor driving an exciter generator. Necessary starter devices in accordance with MIL-C-2212 shall be provided for the motor.

3.4.2 Accessory exciter. An exciter driven as an accessory shall be in accordance with MIL-G-3111, except as otherwise specified herein.

3.4.3 Insulation. Insulation shall be class B or H of MIL-E-917.

3.4.4 Enclosures. Enclosures shall be dripproof-protected in accordance with MIL-STD-108

3.4.5 Rating of generator. Rating shall be as required to provide excitation and field forcing for the minesweeping generator.

3.4.6 Stray magnetic fields. Extreme care shall be taken in the design of exciter generators to minimize stray magnetic fields. Exciter generators shall have not less than six main field poles and the same number of interpoles.

3.5 Minesweeping control equipment.

3.5.1 General. Minesweeping control equipment shall consist of a switchboard, contactor panel, and synchronizer signal generator to be located in the engine room, a safe-run switch to be located at the terminal box, and a transfer panel and Mark VII auxiliary controller to be located in the pilot house. Mark VII auxiliary controller is not covered by this specification.

3.5.1.1 Insulation. Insulation shall be class B except where class A is authorized by the Naval Ship Engineering Center (NAVSEC).

3.5.2 Switchboard, transfer panel and contactor panel.

3.5.2.1 Switchboard. Switchboard shall be a dripproof, nonmagnetic steel enclosure for deck mounting. Switchboard shall contain necessary apparatus for control of the minesweeping generator and exciter motor-generator set and for control of the 500 Hz signal source. A voltmeter, ammeters, a rheostat within the bridge circuit for control of the generator output, a deperm control rheostat, a rheostat for balance of forward and reverse pulses, a control selector switch, a ground test switch, ground detector lights, a control power switch, and start-stop push buttons for the exciter motor-generator set shall be mounted deadfront on the cabinet door. Relays, resistors, transformers, magnetic amplifiers if used, and 500 Hz signal tuning capacitors shall be mounted within the cabinet. Adjustable resistors shall be of the deadfront, screwdriver adjustable type. A relay shall be provided for keying the 500 Hz synchronizing-signal. The "Forward" and "Reverse" relays shall have

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contacts for operating the polarity lights. Internal apparatus except the ammeters shall be wired to terminal strips located in the upper portion of the switchboard. A hand rail shall be provided on the front of the switchboard. A gasketed, undrilled lead plate of ample size for entrance of control cables shall be provided on the top of the switchboard. Rheostats shall have clockwise rotation to increase voltage and shall have stops to prevent brush from breaking contact with the resistance element.

3.5.2.2 Transfer panel. Transfer panel shall contain an automatic-off manual transfer switch, an impulse switch, a start-stop switch, an on-off switch for field indicator lights, and a load ammeter and voltmeter. Impulse switch shall provide spring return to the "OFF" position for normal operation, but shall be provided with a locking feature to hold it in the "Forward" or "Reverse" position when necessary. Panel shall be a dripproof enclosure, and shall be designed for bulkhead mounting. Gasketed lead plates shall be provided for entrance by cables at either top or bottom.

3.5.2.3 Contactor panel. Contactor panel shall be a dripproof nonmetallic steel enclosure for deck mounting containing an electrically or pneumatically operated contactor to disconnect one side of the generator from the line and to connect the 500 Hz synchronizer signal source to the line during "OFF" time. Contactor is not required to break load current.

3.5.2.4 Meters. Meters shall be 4-1/2 inches rectangular flange, flush mounting, 250 degrees nominal scale, white dial background, in accordance with MIL-M-16034.

3.5.3 Synchronizer signal source. Synchronizer signal source shall be a static inverter with 450 V, 3-phase, 60 Hz input, and 23/28.5/34 V, single phase, 500 Hz, 1 kW output. Parts shall be standard parts as specified in MIL-STD-242. If nonstandard parts are proposed, data shall be submitted to NAVSEC in accordance with MIL-STD-749. The synchronizer signal source shall be located in the contactor panel and shall be used for no other purpose than the synchronizing signal.

3.6 Gas turbine and reduction gear. Gas turbine engine and reduction gear shall conform to type A of MIL-E-17341, except as otherwise specified herein. Rotation may be either hand.

3.6.1 Gas turbine compressor and exhaust ducting shall not be the responsibility of the engine builder beyond the engine ducting mounting flange. However, the engine builder shall provide guidance to the shipbuilder, if requested, in the design of the intake and exhaust duct arrangements. Eductor and flange dimensions shall be provided on the installation drawings.

3.6.2 Means shall be provided for properly securing silencing components to the unit. The furnishing and installation of the silencing components shall be the responsibility of the shipbuilder. If additional noise reduction is required in the installation onboard ship, the engine builder shall provide recommendations.

3.6.3 Engine safety devices.

3.6.3.1 A turbine gas temperature limiter with a fuel limiting device shall be provided to prevent operation at temperatures in excess of 75°F above that temperature required to develop rated power to meet the turbine design life.

3.6.3.2 Engine shall be provided with a speed limiting device which will prevent generator speed in excess of 107 percent of design speed.

3.6.4 Starting. Starting shall be accomplished by hydraulic motor.

3.6.5 Accessories. Engine shall be complete in all respects and shall include the following components and accessories:

(a) Main reduction and accessory gears.
(b) Fuel boost pump, governor unit, and overspeed trip.
(c) Lubricating oil filter.
(d) Lubricating oil pump.
(e) Lubricating oil cooler.
(f) Fuel filter.
(g) Igniter system complete (including plugs and coils).
(h) Engine mounts.
(i) Engine gas temperature measuring thermocouples.
(j) Turbines shaft(s) tachometer(s).
(k) Solenoid fuel shut-off between the governor and fuel nozzle.

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(l) Generator bearing lube oil scavenging pump (if required).
(m) Instruments and instrument panel.
(n) Sea water pump(s), including pump for generator cooling.
(o) Engine exhaust eductor system design.
(p) Nonmagnetic flywheel of the required size.
(q) Starter motor.

3.6.6 Gas turbine electrical control equipment. Wiring and electrical equipment shall be shielded from interaction with stray electromagnetic signals and shall be designed to avoid producing such radiation. Interference shielding shall be provided in accordance with MIL-I-16165.

3.6.7 Pumps and piping. Pumps, except fuel boost and auxiliary lube oil pump (if used), shall be driven by the generator set as attached accessories. Piping between attached accessories and the engine shall be provided. Piping connections for connection to the ships system shall be provided with flanges in accordance with MIL-F-20670. Pumps shall have adequate capacity for unit operation under the conditions specified herein. If required, electrically driven fuel boost and auxiliary lube oil pumps shall be supplied. The sea water pump suction head and dry pipe suction requirements shall be as specified (see 6.2.1).

3.6.8 Lubricating oil system. Unit shall be provided with a lube oil reservoir. Lube oil system shall provide the required lubrication for engine, gear, and generator bearings requiring oil lubrication. Lube oil pressure controls, as required for the turbine and generator set, shall be furnished. Lube oil sump capacity shall be sufficient to allow the set to operate at rated pulsing load for a period of not less than 8 hours without adding oil to the sump. An auxiliary motor driven pump shall be provided if needed for proper lubrication during start-up or run-down.

3.6.9 Oil cooler. An oil cooler, using sea water as a cooling medium and conforming with type A, class 3 of MIL-C-15730, shall be furnished with each unit. Cooling capacity shall be determined on the basis of sea water at 95°F. If synthetic oil is used, no material containing more than 5 percent copper shall be used in the cooler lube system.

3.6.10 Water pumps. Sea water pump(s) shall be of the centrifugal type.

3.6.11 Governing system. Engine shall be provided with a governing system to provide the electrical performance requirements specified in 3.3.1 and 3.3.2 and an overspeed shutdown device.

3.6.12 Overspeed governor. Overspeed governor shall automatically shut-off the fuel to the turbine when the speed reaches 107-110 percent of rated speed.

3.6.13 Remote shutdown provisions. Means shall be provided for emergency shutting down of the turbine from a remote station. (The necessary connections between the remote station and the shut-off device on the turbine shall be provided by the activity installing the generator set.)

3.6.14 Turbine control panel(s). Turbine control panel or panels shall be of dripproof-protected construction. These panel(s) shall be designed for bulkhead mounting. Gages, thermometers, instruments, or indicators on the control panels shall be of the flush-mounted type and shall include, but shall not be limited to, the following:

(a) Sea water pump discharge pressure gage.
(b) Lubricating oil pump discharge pressure gage.
(c) Generator lube oil pressure gage, (if oil lubricated bearing is used).
(d) Temperature indicator for measuring lube oil temperature in °F in and out of the cooler.
(e) Temperature indicator for measuring temperature of the turbine gas, °F.
(f) Indicator showing gas producer and power turbine speed, revolutions per minute (r/min).
(g) Speed control switch (if governor synchronizing motor is used).
(h) Stop-start switch for manually starting or stopping the unit.
(i) Indicator and warning lights.
(j) Various reset devices (as applicable).
(k) Various terminal boards (as applicable).
(l) Any other appurtenances necessary for proper operation and control of the engine and generator set.

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3.7 Inertia flywheel. Inertia flywheel shall be made of nonmagnetic material having an electrical conductivity less than 10 percent of that of copper.

3.8 Flywheel guard. Flywheel and flange shall be provided with a guard. Guard shall be designed to prevent injury to personnel who may be thrown against it in a sea way and prevent objects larger than 1/2 inch in diameter from coming in contact with the flywheel or shafting.

3.9 Reliability and maintainability requirements.

3.9.1 Reliability and maintainability quantitative requirements. The specified mean-time-between-failure (MTBF) (θ_o) of the minesweeping power system shall be at least 3,000 hours. Equipment repair time (ERT), shall be 3 hours. Maximum repair time for any item shall not exceed 16 hours.

3.9.1.1 Failure definition. Equipment failure is defined as a condition resulting in an unavoidable, forced shutdown of the equipment, that cannot be restored without some degree of disassembly to repair or replace an internal element.

3.9.2 Reliability and maintainability program plan. The contractor shall develop, maintain, and implement a reliability and maintainability program plan.

3.9.3 Reliability prediction. The contractor shall provide a reliability prediction in which the failure rate data and their sources shall be fully documented. For parts where no failure rate is available, a failure rate shall be estimated and the basis for the estimat shall be stated in the calculations. Derating factors used for each part and the stress levels at which each part is operating shall be stated in reliability calculations. The stress levels shall be those as determined from test insofar as practicable. In those instances, where test data is not available design data shall be used.

3.9.4 Failure mode and effects analysis. The contractor shall perform a failure mode and effects analysis (FMEA) which shall be conducted to the lowest replaceable unit level.

3.9.5 Maintainability prediction. The contractor shall perform a maintainability analysis in which the failure rates used shall be those which have been determined in the reliability predictions (see 3.9.3).

3.9.6 Maintenance engineering analysis. The contractor shall perform a maintenance engineering analysis (MEA) for each functional system, equipment, unit, assembly or sub-assembly which requires preventative or corrective maintenance, and the continued operation of which is essential to the successful completion of the mission itself. The MEA shall include, as a minimum, the completion of worksheets III and IV of MIL-M-24365.

3.10 Repair parts and special tools. When specified (see 6.2.1), repair parts shall be furnished in accordance with MIL-STD-1552 and MIL-STD-1561.

3.10.1 Onboard repair parts for turbine, reduction gear, and turbine accessories shall be in accordance with MIL-E-17341.

3.10.2 Onboard repair parts for electrical equipment shall be in accordance with table

TABLE I. Repair parts for electrical equipment.

Items	Percent of installed items, per component, to be furnished as repair parts
Bearings	100
Brushes	200
Brush holders	50
Brush springs	100
Capacitors (each type and size)	10
Couplings, flexible (if used)	100
Filters, choke	10
Fuses, standard	50
Fuses, special	300
Indicator light lamps, special	300
Indicator light lamps, standard	200
Indicator light lenses	100

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TABLE I. Repair parts for electrical equipment. (Continued)

Items	Percent of installed items, per component, to be furnished as repair parts
Indicator light sockets	50
Magnetic amplifier rectifiers	100
Relay coils (each type and size)	50
Relay contacts and contact springs	100
Resistors (each type and size)	50
Rheostats (each type and size)	25
Semiconductors (each type and size)	50
Shaft seals	100
Switch contacts and contact springs	100
Switch handles (each type and size)	10
Switches with nonrenewable contacts	100
Transformers (each type and size)	10
Zincs, electrolytic (each type and size)	200
Plugs, tube (phenolic)	50

3.10.3 Special tools. Special tools for the assembly, disassembly, and maintenance of the turbine, gear, generator, and associated equipment, shall be furnished as a part of the generator set. Special tools are defined as those tools not listed in the National Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Administration Service (DCAS)).

3.11 Technical data. The contractor shall prepare technical data in accordance with the data ordering documents included in the contract or order (see 6.2.2) for the following, and as specified in 3.11.1 through 3.11.2.3:

(a) Reliability and maintainability program plans (see 3.9.2).
(b) Reliability prediction report (see 3.9.3).
(c) Failure mode and effects analysis report (see 3.9.4).
(d) Maintainability prediction report (see 3.9.5).
(e) Maintainance engineering analysis report (see 3.9.6).

3.11.1 Drawings. The contractor shall prepare drawings in accordance with the data ordering document included in the contract or order (see 6.2.2).

3.11.2 Manuals. In addition to the general requirements covered by the data ordering document (see 6.2.2), the features specified in 3.11.2.1 through 3.11.2.3 shall be included.

3.11.2.1 Separate manuals (or separate parts in one manual) shall be furnished for the engine (including reduction gear, flywheel, pedestal bearing, engine accessories, controls and subbase) and for the minesweeping electrical equipment. The manual for the minesweeping electrical equipment shall include coverage of the Mark VII auxiliary controller (see 3.5.1).

3.11.2.2 Engine manual. Engine manual shall be unclassified and shall include graph(s) or tabulation(s) relating electrical output to compressor inlet air temperature, tabulated results of combined unit tests and reduced installation drawing(s).

3.11.2.3 Electrical manual. Electrical manual shall be unclassified and shall include generator design ratings and engine-generator ratings for the 5-1/2 second "ON", 9-1/2 second "OFF" duty cycle, and for continuous duty, both related to compressor inlet air temperature. In addition, the electrical manual shall have a separate confidential supplement containing a graph or graphs from which, with any one of the following parameters established: (a) pulse current, (b) "ON"-time, (c) peak kW, the maximum permissible operating conditions of: (a) maximum pulse current, (b) maximum "ON"-time, (c) maximum peak kW, (d) minimum "OFF"-time, all related to compressor inlet air temperature can be determined.

3.12 Workmanship. Particular attention shall be paid to cleaning of parts prior to assembly and during fabrication. Mold marks, casting sand, and similar conditions shall be removed. Faying surfaces of metal parts shall be machined, bolt holes shall be deburred, and spot faced to assure proper seating of bolts and nuts except that bolt holes in nonstructural parts such as duct work flanges, sheet metal housing, etc., need not be spot faced or machined. Internal wiring of switchboards and panels shall be properly laced and secured to sides or tops of enclosures. Piping and tubing runs shall be as neat and direct as possible and shall be firmly secured to adjacent structure. No loose or unsupported wiring, tubing, or control cables will be permitted.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality program plan. The contractor shall provide and maintain a quality program plan in accordance with the data ordering document included in the contract or order (see 6.2.2).

4.2 General examination. Each piece of equipment shall be examined to ascertain that the material, workmanship, and design are in conformity with specified requirements. Fit of parts shall be observed with particular reference to the interchangeability of such parts as are likely to require replacement during the life of the equipment.

4.3 Generator tests. The first generator of a contract or order shall be tested in accordance with the periodic test schedule of MIL-G-3111 and subsequent generators shall be tested in accordance with the routine test schedule of MIL-G-3111 except as otherwise specified herein. Radio noise and voltage regulation tests will not be required. The ratings specified in 3.3.1 shall be used as the basis for periodic and routine tests.

4.3.1 Heating test. The first generator of a contract or order shall be operated at rated load on a pulse duty cycle of 5-1/2 seconds "ON", 9-1/2 seconds "OFF", or at a continuous root mean square (rms) equivalent of this load, until steady final temperatures have been attained in all parts of the generator. To shorten the heating period, reasonable overloads of current during the preliminary period are permissible. During the progress of the heat run, temperature measurements shall be taken as follows:

(a) Ambient air (measured in accordance with MIL-G-3111).
(b) Air out of generator.
(c) Air out of cooler.
(d) Water into cooler.
(e) Water out of cooler.
(f) Bearing temperatures.
(g) Each end and center of two main field coils.
(h) Each end and center of two commutating pole coils.
(i) Outside of frame.
(j) Main field (by resistance).

4.3.1.1 Immediately on shut-down at the end of the heat run, temperatures shall be taken at the following points: (The machine shall be shut-down and thermocouples or preheated thermometers installed within 3 minutes of the time run is ended. Temperatures shall be recorded at intervals to obtain the cooling curve and the curve plotted and extrapolated back to the instant of ending the run.)

(a) Commutator (at three axially equidistant points).
(b) Risers.
(c) Armature-end connections (front and back).
(d) Armature iron below winding (front and back).
(e) Main and commutating windings.

4.3.1.2 The temperatures recorded during the run and at shut-down shall be compared with temperatures excepted to give the designed life of the insulation. Any temperatures considered exceptionally high shall be reported to NAVSEC for decision as to their acceptability.

4.3.2 Vibration test. Generator tests shall demonstrate conformance with 3.3.12 with the generator rigidly mounted on a test platten.

4.4 Exciter set tests. The first exciter set of each contract or order shall be tested in accordance with the periodic test schedule, and subsequent sets shall be tested in accordance with the routine test schedule of MIL-M-2130 (MIL-G-3111 in the case of an accessory exciter).

4.4.1 Additional tests. Additional tests as specified in 4.4.1.1 through 4.4.1.3 shall be made on the first exciter set.

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4.4.1.1 Efficiency. Conventional efficiency shall be determined for each unit of the set. Efficiencies shall be corrected to a reference temperature of 75°C; however, the test may be made at any ambient temperature above 10°C.

4.4.1.2 Insulation test. An insulation test in accordance with MIL-G-3111 shall be made on each unit of the motor-generator set.

4.4.1.3 Heating test. At the close of the 4-hour full load test, temperature measurements shall be made at windings, bearings, commutators, armature iron, and outside of frames of each unit. The temperatures obtained shall be compared with temperatures expected to give the designed life of the insulation.

4.5 Switchboard contactor panel and transfer panel tests. A switchboard contactor panel and transfer panel shall be used for control of generator power during the period of combined engine-generator tests. Operation over the duration of this period shall constitute the performance tests for these units.

4.6 Combined unit tests. Generator and its prime mover shall be assembled and subjected to the combined unit tests specified herein. During the runs, accurate data shall be taken to show fuel and lubricating oil consumption for each of the power conditions under study.

4.6.1 First unit tests. The first engine-generator set of a new or changed design shall be operated on the ten 7-hour runs specified in table II under the control of its associated exciter set and control apparatus. Each 7-hour run shall be uninterrupted. In the event of any failure during the specified ten runs which necessitates a design change, the entire series of test runs shall be remade after corrections satisfactory to NAVSEC have been made. In the event of interruption of a run, that run shall be remade and the cause of the interruption shall be recorded.

TABLE II. Combined unit test for 1750 kW set.

Run	Amperes	Volts	Wave form	"ON" forward seconds	"OFF" seconds	"ON" reverse seconds	"OFF" seconds	Hours of run
1	5000	350	Square	5-1/2	9-1/2	5-1/2	9-1/2	7
2	2500	175	Square	5-1/2	9-1/2	5-1/2	9-1/2	7
3	1000	70	Square	5-1/2	9-1/2	5-1/2	9-1/2	7
4	500	35	Square	5-1/2	9-1/2	5-1/2	9-1/2	7
5	5000	350	Sine (8 second half cycle time)	8	10	8	10	7
6	5000	350	Saw tooth (8 second half cycle time)	8	10	8	10	7
7	5000	350	Sine (4 second half cycle time)	8	10	8	10	7
8	2500	175	Sine (4 second half cycle time)	8	10	8	10	7
9	3150	220	Square	5	0	5	0	7
10	1000	70	Square	5	0	5	0	7

4.6.1.1 The first gas turbine shall pass the applicable tests for type A engines specified in MIL-E-17341 under acceptance requirements.

4.6.1.2 Shock test. Shock tests shall be conducted to assure compliance with 3.2.2.

4.6.2 Subsequent unit tests. Subsequent engine-generator sets of identical design submitted for acceptance within 1 year of test of the last previous unit shall be operated on the three 7-hour runs, numbers 1, 3, and 9, of table II. Subsequent engine-generator sets of identical design submitted later than 1 year from test of the last previous unit shall be operated on the full schedule of ten 7-hour runs.

4.6.3 Oscillograph records. An oscillograph record of the line current of each generator, line voltage, excitation voltage, excitation current, and generator speed taken simultaneously shall be made during a part of each of the required test runs. Each oscillograph record shall have a timing reference and calibration marks from which time and magnitude variations may be accurately determined. Oscillograph records shall be forwarded to NAVSEC as a part of the test reports (see 4.18).

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4.7 Starting test. The first engine-generator set shall be started, by the method and apparatus intended for shipboard installation, 20 times from standstill condition. Two false starts during the test shall require rerun of the starting test. Subsequent engine-generator sets shall be started 10 times, and one false start shall require rerun of the starting test.

4.8 Acceleration test. Engine-generator sets shall be started from prevailing ambient conditions and brought to full normal rated generator speed. Four such runs shall be made and the average time shall be designated as "starting time".

4.9 Overspeed test. An overspeed test shall be conducted on each engine-generator set to demonstrate that no damage will occur under the following overspeed conditions over a time interval of 5 minutes:

(a) Single shaft engines - 105 percent of generator rated speed.
(b) Two shaft engines - 110 percent of generator rated speed.

4.10 Safety test. Safety devices incorporated in the unit shall be tested at least three times during actual operation of the engine to demonstrate their effectiveness. No damage shall result from this operation.

4.11 Test conditions. Fuel and lubricating oil used during the tests shall be in accordance with MIL-E-17341. Temperature of water to the sea water pumps shall be corrected to 95°F. Two or more tests may be combined provided objectives of each test are achieved as effectively as though separate tests were made.

NOTE: Rated load is defined as the maximum power level at which the engine shall be required to operate when driving the generator at rated pulse load as specified herein. Rated pulse load is defined as that load imposed upon the engine(s) when the generator is delivering 1750 kW at 5-1/2 seconds on and 9-1/2 seconds off square wave pulsing.

4.12 Inclination test. An inclination test of the complete engine-generator set is not required. However, the contractor shall furnish sufficient evidence to NAVSEC showing that the complete generator set has been designed to operate successfully in accordance with the requirements of 3.2.5.

4.13 Vibration test. A vibration test shall be conducted on each generator set in accordance with MIL-STD-167-1, type I to assure compliance with 3.2.12.

4.14 Reliability demonstration test. The reliability demonstration test shall be conducted in accordance with test plan VI of MIL-STD-781. A failure is defined as the occurrence of any of the following:

(a) Failure to operate.
(b) Output parameters outside of specified limits.
(c) Requirement for maintenance during test not specifically identified as scheduled (preventative) maintenance prior to initiation of test.

4.15 Maintainability demonstration test. The maintainability demonstration test shall be conducted in accordance with method IV of MIL-STD-471. All preventative maintenance tasks shall also be demonstrated.

4.16 Rejection criteria. If a reject decision is reached during the reliability and maintainability tests, the contractor shall perform necessary corrective action. Verification of the modifications will be accomplished by complete retest or partial retest as determined by NAVSEC.

4.17 Reliability and maintainability data collection and analysis. The contractor shall conduct a reliability and maintainability data collection, analysis, and corrective action program throughout all phases of testing. Where a failure is experienced or discovered during factory examination and tests, a failure analysis as defined in MIL-STD-721 and a failure/malfunction report shall be prepared.

4.18 Test procedures and reports. The contractor shall prepare test procedures and reports in accordance with the data ordering documents included in the contract or order (see 6.2.2).

4.19 Inspection of preparation for delivery. The preservation-packaging, packing, and marking shall be inspected for compliance with section 5 of this document.

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5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.3.)

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.2.1).

5.1.1 Level A. Unless otherwise specified herein, cleaning, drying, preservation-packaging, and cushioning shall be in accordance with MIL-P-116, and selection of the sub-method under a particular method of preservation (unit protection) shall be at the contractor's option.

5.1.1.1 Generator set. Each generator set (see 3.1) shall be unit protected by method IIa. The flexible barrier shall be provided with a humidity indicator plug in lieu of the 3-spot humidity indicator card.

5.1.1.1.1 Basic engine unit. Basic engine unit including driven equipment shall be cleaned, dried, and preserved as follows:

(a) Gas turbine engine shall be cleaned, dried, and preserved in accordance with MIL-E-5607. Upon completion of preservation of the lubricating and fuel oil systems, the throw-away elements shall be replaced with new elements.
(b) Reduction gear, accessory gears, and couplings shall be cleaned and preserved in accordance with MIL-P-17286.
(c) Coolers shall be cleaned and preserved in accordance with MIL-C-15730. External unpainted ferrous surfaces shall be coated with a preservative conforming to type P-1 or type P-19 of MIL-P-116.
(d) Water pumps shall be cleaned and preserved as follows:
(1) Pumps shall be cleaned in accordance with process C-1 of MIL-P-116. Internal surfaces coming into contact with water shall be flushed or fog sprayed with a preservative conforming to type P-3 of MIL-P-116, or when chemical boil-out cannot be used for preservative removal, or where removal by hot water or steam is desired, grade 5 of MIL-C-16173 shall be used. All external unpainted ferrous surfaces shall be coated with a preservative conforming to type P-1 or type P-19 of MIL-P-116.

5.1.1.1.2 Generator. Generator shall be cleaned, dried, and preserved in accordance with MIL-E-16298.

5.1.1.2 Associated equipment, accessories, and instruments that are detached, repair parts and tools shall be cleaned, dried, and unit protected in accordance with the following:

(a) MIL-P-17286 and MIL-R-196 for mechanical material.
(b) MIL-E-17555 and MIL-E-16298 for electrical/electronic equipment.

Repair parts and tools unless used in sets, pairs, or quantities greater than one shall be individually preserved and packaged.

5.1.1.2.1 Semiconductors or solid state devices. Semiconductors such as diodes, transistors, integrated circuits as well as circuit boards or chassis, in which they are incorporated, shall be individually packaged in a barrier bag conforming to class E, style 1, type I or II, or class F, style 1, type 1 of MIL-B-117. MIL-B-117 bag material shall employ aluminum foil as a laminate of the bag barrier material. Bag closure shall be effected by heat sealing. Leads and terminals shall be protected from damage by means of the container (carrier) design, die cut inserts, or by the use of noncorrosive cushioning material. Leads and other projecting parts may be used for positioning, but shall not be subjected to loads or other stresses such as bending or twisting that can damage the entry seals. For level C preservation-packaging, semiconductors or solid state devices subject to electromagnetic degradation shall be protected with a wrap of aluminum foil or a barrier bag employing aluminum foil as a laminate of the bag material. Leads and terminals shall be protected as specified herein.

5.1.1.3 Repair parts boxes. When specified (see 6.2.1), repair parts and tools shall be packed in boxes conforming to MIL-B-233, type W, class at the option of the contractor.

5.1.1.3.1 Index list. An index list shall be inserted in each shipping container listing repair parts and tools accompanying the equipment or procured as a set or kit. The

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list shall be placed in the index list support located on the interior side of the cover of the repair parts box or placed in the inside of the box for quick accessibility. The list shall completely itemize the contents of the container.

5.1.1.3.2 Cushioning, dunnage, and wrapping materials.

5.1.1.3.2.1 Level A preservation-packaging, level A and B packing. Use of all types of loose-fill materials for applications such as cushioning, filler, stuffing, and dunnage for material destined for shipboard use is prohibited.

5.1.1.3.2.2 Level C preservation-packaging and packing. Unless otherwise specified (see 6.2.1), use of all types of loose-fill materials for applications such as cushioning, filler, stuffing, and dunnage for material destined for shipboard use is prohibited. When approved for use by the contract or order, unit packages and containers (interior and exterior) shall be marked or labeled as follows:

"CAUTION

Contents cushioned etc. with loose-fill material.
Not to be taken onboard ship.
Remove and discard loose-fill material before shipboard storage.
If required, recushion with cellulosic material bound fiber,
fiberboard, or transparent flexible cellular material."

5.1.2 Level C. Generator sets, associated equipment, accessories, and instruments that are detached, repair parts and tools shall be cleaned, dried, preserved and packaged in accordance with the contractor's normal procedures or alternate means that shall assure protection against corrosion, deterioration, and damage during shipment from the manufacturer's plant to the using activity and for a period of approximately 6 months thereafter.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2.1).

5.2.1 Level A and B.

5.2.1.1 Generator set. Each generator set shall be anchored, blocked, braced, and cushioned (see 5.4) and packed in a plywood sheathed crate conforming to MIL-C-104, type II, class 2, style a or b at the contractor's option. Use of plywood in lieu of lumber sheathing for crate top construction is acceptable. Crates shall be provided with an examination port for use in visually viewing the humidity indicator mounted in the barrier material.

5.2.1.2 Associated equipment, detached accessories and instruments, repair parts and tools shall be packed in containers as specified in MIL-E-17555. Rough handling test(s) as specified in MIL-E-17555 is not required. Repair parts boxes (see 5.1.1.3) when required, shall be packed in accordance with MIL-P-17286.

5.2.2 Level C. Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the first using activity for early installation in accordance with Uniform Freight or National Motor Freight Classification Rules or other carrier regulations as applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required (see 6.2.1), interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.4 Anchoring, blocking, bracing, and cushioning. Anchoring, blocking, bracing, and cushioning shall be in accordance with MIL-STD-1186 and the applicable commodity or container specification and appendix thereto.

5.5 Drawings. Drawings shall be prepared for shipment in accordance with the requirements of the applicable specification.

5.6 Technical manuals. Technical manuals, which accompany shipments that are packed level A or B, shall be packaged in a transparent waterproof plastic bag, minimum 4 mil thick. Closure shall be by heat sealing. Technical manuals shall not be placed within any flexible sealed barrier enclosing components. The copy(s) of the manual shall be placed in the shipping container housing the main unit. Packing lists shall indicate which container contains the technical manual(s) and shall also state the approximate location therein. For ease of removability the location of the manual shall be such that it is readily accessible when the container is opened. Technical manuals, when shipped in bulk quantities, shall not be individually wrapped, but shall be packed in accordance with the requirements of the applicable technical manual specification or packed in containers conforming to the requirements for level A, B, or C, as specified (see 6.2.1).

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6. NOTES

6.1 Intended use. The generator set is intended to supply current to a ship's magnetic minesweeping cable.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Sea water pump suction head requirements (see 3.6.7).
(c) When repair parts are required (see 3.10).
(d) Level of preservation-packaging and packing required (see 5.1, 5.2, and 5.6).
(e) Repair parts boxes when required (see 5.1.1.3).
(f) When loose-fill cushioning material is permitted (see 5.1.1.3.2.2).
(g) Special marking required (see 5.3).

6.2.2 Data requirements. When this specification is used in a procurement which invokes the provision of the "Requirements for Data" of the Armed Services Procurement Regulations (ASPR), the data identified below, which are required to be developed by the contractor, as specified on an approved Data Item Description (DD Form 1664), and which are required to be delivered to the Government, should be selected and specified on the approved Contract Data Requirement List (DD Form 1423) and incorporated in the contract. When the provisions of the "Requirements for Data" of the ASPR are not invoked in a procurement, the data required to be developed by the contractor and required to be delivered to the Government should be selected from the list below and specified in the contract.

Paragraph	Data requirements	Applicable DID	Options
3.9.2	Reliability program plan	DI-R-2113	-------
3.9.2	Maintainability program plan	UDI-R-23558	-------
3.9.3	Reliability prediction report	DI-R-2117	-------
3.9.4	Failure mode and effects analysis report	DI-R-2115	-------
3.9.5	Maintainability prediction report	DI-R-2128	-------
3.9.6	Maintenance engineering analysis report (MEA)	UDI-L-26377	-------
3.11.1	Drawings, engineering and associated lists, level 3 (production)	DI-E-7015	Design activity designation - Contractor Design activity drawing numbers - Contractor Delivery of hard copy - Procuring activity
3.11.2	Manual, technical, preliminary	DI-M-2043	MIL-M-15071, type I
3.11.2	Manual, technical, standard, basic issue	DI-M-2044	MIL-M-15071, type I
4.1.1	Quality program plan	UDI-R-23743	-------
4.17	Failure/malfunction report	UDI-T-23724	-------
4.18	Test procedures	UDI-T-23732	-------
4.18	Test/inspection report	UDI-T-23473	-------

(Copies of data item descriptions required by the contractors in connection with specifi procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.)

6.2.2.1 The data requirements of 6.2.2 and any task in section 3, 4, or 5 of the specification required to be performed to meet a data requirement may be waived by the procuring/purchasing activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item procured to this specification. This does not apply to specific data which may be required for each procurement regardless of whether an identical item has been supplied previously (for example, test reports).

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6.3 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are procured by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.4 Changes from previous issue. The symbol "#" is not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity
Navy - SH
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